Brutal Series B23 - Electric Current / Soil / Fractions & Decimals / Triangles

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. An electric heater rated 2000 watts runs for 3.5 hours each day for 4 days. If 1 unit is 1000 watt-hours and each unit costs 6 rupees, what is the total cost? [Electric Current & its Effects]

- **(A)** 120 rupees
- **(B)** 84 rupees
- **(C)** 168 rupees
- **(D)** 336 rupees
- **(E)** 210 rupees
- **(F)** 150 rupees
- **(G)** 42 rupees

2. In a percolation test, 200 millilitres of water pass through a soil sample in 40 minutes. The percolation rate is the amount divided by the time. What is the rate, and is this soil more like sandy or clayey? [Soil]

- **(A)** 2 mL/min, like clayey soil
- **(B)** 40 mL/min, like sandy soil
- **(C)** 5 mL/min, like clayey soil
- **(D)** 5 mL/min, like only loamy soil
- **(E)** 5 mL/min, like sandy soil
- **(F)** 0.2 mL/min, like sandy soil

3. A tank is 4/5 full. After 1/3 of the water now in it is used, what fraction of the FULL tank still remains? [Fractions & Decimals]

- **(A)** 8/15
- **(B)** 8/5
- **(C)** 11/15
- **(D)** 12/15
- **(E)** 2/3
- **(F)** 7/15

4. In a triangle the second angle is twice the first, and the third is 30 degrees more than the first. Find the three angles. [The Triangle & its Properties]

- **(A)** 40, 80 and 60 degrees
- **(B)** 50, 100 and 30 degrees
- **(C)** 37.5, 75 and 67.5 degrees
- **(D)** 30, 60 and 90 degrees
- **(E)** 45, 90 and 45 degrees
- **(F)** 36, 72 and 72 degrees
- **(G)** 37.5, 70 and 72.5 degrees

5. A circuit needs about 6 volts. A child has only 1.5-volt cells joined in series so the voltages add. How many cells give exactly 6 volts, and what happens if one of them is reversed? [Electric Current & its Effects]

- **(A)** 4 cells; reversing one drops it to 4.5 volts
- **(B)** 4 cells; reversing one drops it to 1.5 volts
- **(C)** 6 cells; reversing one gives 0 volts
- **(D)** 5 cells; reversing one drops it to 3 volts
- **(E)** 3 cells; reversing one gives 3 volts
- **(F)** 4 cells; reversing one keeps 6 volts
- **(G)** 4 cells; reversing one drops it to 3 volts

6. Soil sample A lets 300 mL of water pass in 25 minutes; soil sample B lets 300 mL pass in 60 minutes. How much FASTER (in mL per minute) does water percolate through A than through B? [Soil]

- **(A)** 0.7 mL/min faster
- **(B)** 5 mL/min faster
- **(C)** 12 mL/min faster
- **(D)** 2 mL/min faster
- **(E)** 7 mL/min faster
- **(F)** 35 mL/min faster
- **(G)** 17 mL/min faster

7. A pen costs 12.50 rupees. A box has 8 pens, and the shop gives a discount of 0.50 rupee on each pen. What is the cost of a full box after the discount? [Fractions & Decimals]

- **(A)** 99.50 rupees
- **(B)** 95.50 rupees
- **(C)** 96 rupees
- **(D)** 96.50 rupees
- **(E)** 104 rupees
- **(F)** 12 rupees
- **(G)** 100 rupees

8. An exterior angle of a triangle measures 120 degrees, and one of its two remote interior angles is 50 degrees. Find the other remote interior angle and the interior angle next to the exterior one. [The Triangle & its Properties]

- **(A)** 50 degrees; the adjacent interior angle is 60 degrees
- **(B)** 60 degrees; the adjacent interior angle is 60 degrees
- **(C)** 70 degrees; the adjacent interior angle is 60 degrees
- **(D)** 30 degrees; the adjacent interior angle is 60 degrees
- **(E)** 70 degrees; the adjacent interior angle is 70 degrees
- **(F)** 70 degrees; the adjacent interior angle is 120 degrees

9. Three appliances on one socket draw 4 amperes, 6 amperes and 2.5 amperes. A fuse must carry the total current with a 25 percent safety margin above it. What is the smallest whole-ampere fuse to use? [Electric Current & its Effects]

- **(A)** 12 amperes
- **(B)** 14 amperes
- **(C)** 13 amperes
- **(D)** 15 amperes
- **(E)** 16 amperes
- **(F)** 20 amperes
- **(G)** 17 amperes

10. Going downward from the surface, which is the correct order of the soil layers? [Soil]

- **(A)** topsoil, then subsoil, then weathered rock, then bedrock
- **(B)** topsoil, then bedrock, then subsoil, then weathered rock
- **(C)** bedrock, then weathered rock, then subsoil, then topsoil
- **(D)** bedrock, then subsoil, then topsoil, then weathered rock
- **(E)** weathered rock, then topsoil, then subsoil, then bedrock
- **(F)** topsoil, then subsoil, then bedrock, then weathered rock
- **(G)** subsoil, then topsoil, then weathered rock, then bedrock

11. How many 3/4-metre ribbons can be cut from a 9-metre roll, and how much ribbon is left over? [Fractions & Decimals]

- **(A)** 12 ribbons, with none left over
- **(B)** 9 ribbons, with 3/4 m left over
- **(C)** 12 ribbons, with 1/2 m left over
- **(D)** 16 ribbons, with none left over
- **(E)** 11 ribbons, with 1/4 m left over
- **(F)** 12 ribbons, with 3/4 m left over
- **(G)** 6 ribbons, with none left over

12. The two legs of a right-angled triangle measure 8 cm and 15 cm. Work out the hypotenuse and the perimeter. [The Triangle & its Properties]

- **(A)** hypotenuse 17 cm; perimeter 40 cm
- **(B)** hypotenuse 289 cm; perimeter 312 cm
- **(C)** hypotenuse 23 cm; perimeter 46 cm
- **(D)** hypotenuse 17 cm; perimeter 30 cm
- **(E)** hypotenuse 17 cm; perimeter 23 cm
- **(F)** hypotenuse 12 cm; perimeter 35 cm
- **(G)** hypotenuse 13 cm; perimeter 36 cm

13. A heating coil produces 150 joules of heat every second. How long does it take to supply 9000 joules, and how much heat does it give in 2 minutes of running? [Electric Current & its Effects]

- **(A)** 45 seconds; 18000 joules in 2 minutes
- **(B)** 60 seconds; 300 joules in 2 minutes
- **(C)** 60 seconds; 9000 joules in 2 minutes again
- **(D)** 120 seconds; 18000 joules in 2 minutes
- **(E)** 90 seconds; 18000 joules in 2 minutes
- **(F)** 60 seconds; 9000 joules in 2 minutes
- **(G)** 60 seconds; 18000 joules in 2 minutes

14. Arranged from the LARGEST to the smallest particles, the soils are ___ , which is also the order from fastest to slowest percolation. [Soil]

- **(A)** sandy, then loamy, then clayey
- **(B)** silty, then sandy, then clayey
- **(C)** clayey, then loamy, then sandy
- **(D)** loamy, then sandy, then clayey
- **(E)** sandy, then clayey, then loamy
- **(F)** loamy, then clayey, then sandy
- **(G)** clayey, then sandy, then loamy

15. Which of these is the LARGEST: 5/8, 7/12, 0.6, or 3/5? [Fractions & Decimals]

- **(A)** 7/12 and 5/8 tie for largest
- **(B)** 3/5 and 0.6 tie for largest
- **(C)** 7/12
- **(D)** they are all equal
- **(E)** 3/5
- **(F)** 0.6
- **(G)** 5/8

16. Two sides of a triangle are 7 cm and 4 cm. Between which two whole-number lengths must the third side lie? [The Triangle & its Properties]

- **(A)** from 4 cm to 11 cm
- **(B)** from 7 cm to 11 cm
- **(C)** from 3 cm to 11 cm
- **(D)** from 3 cm to 10 cm
- **(E)** from 1 cm to 11 cm
- **(F)** from 4 cm to 10 cm

17. Coil A has 100 turns of wire and lifts 20 steel clips with a magnet. Keeping the same current, the lifting power grows in step with the number of turns. How many clips should a coil of 250 turns lift? [Electric Current & its Effects]

- **(A)** 70 clips
- **(B)** 60 clips
- **(C)** 100 clips
- **(D)** 50 clips
- **(E)** 45 clips
- **(F)** 25 clips

18. 100 grams of clayey soil holds 45 grams of water, while 100 grams of sandy soil holds only 18 grams. The clayey soil holds what percent MORE water than the sandy soil? [Soil]

- **(A)** 75 percent more
- **(B)** 60 percent more
- **(C)** 2.5 percent more
- **(D)** 150 percent more
- **(E)** 27 percent more
- **(F)** 250 percent more
- **(G)** 40 percent more

19. A recipe needs 2 1/4 cups of flour. To make 3 1/3 times the recipe, how many cups of flour are needed? [Fractions & Decimals]

- **(A)** 7 cups
- **(B)** 6 cups
- **(C)** 8 1/4 cups
- **(D)** 7 1/2 cups
- **(E)** 7 1/3 cups
- **(F)** 5 7/12 cups

20. An isosceles triangle has a vertex (apex) angle of 96 degrees. What is each of its base angles? [The Triangle & its Properties]

- **(A)** 96 degrees each
- **(B)** 42 degrees each
- **(C)** 138 degrees each
- **(D)** 45 degrees each
- **(E)** 84 degrees each
- **(F)** 48 degrees each

21. Two identical bulbs are connected to one cell. Compared with a single bulb alone, in a SERIES connection each bulb is ___ , and if one bulb fuses the other ___ . [Electric Current & its Effects]

- **(A)** brighter; and the other goes off
- **(B)** dimmer; and the other gets brighter
- **(C)** the same; and the other gets brighter
- **(D)** dimmer; and the other stays on
- **(E)** brighter; and the other stays on
- **(F)** dimmer; and the other also goes off
- **(G)** the same; and the other goes off

22. A soil percolates at 4 millilitres per minute. How long will 250 millilitres of water take to pass through, given in minutes and seconds? [Soil]

- **(A)** 1000 minutes
- **(B)** 60 minutes 0 seconds
- **(C)** 62 minutes 25 seconds
- **(D)** 62 minutes 50 seconds
- **(E)** 62 minutes 5 seconds
- **(F)** 62 minutes 30 seconds
- **(G)** 65 minutes 0 seconds

23. From 50, subtract the sum of 12.75, 9.6 and 0.08. What is left? [Fractions & Decimals]

- **(A)** 37.57
- **(B)** 27.57
- **(C)** 27.65
- **(D)** 26.57
- **(E)** 28.57
- **(F)** 22.43

24. An equilateral triangle has the same perimeter as a square of side 9 cm. Find each side of the triangle and each of its angles. [The Triangle & its Properties]

- **(A)** each side 12 cm, each angle 90 degrees
- **(B)** each side 36 cm, each angle 60 degrees
- **(C)** each side 9 cm, each angle 60 degrees
- **(D)** each side 18 cm, each angle 60 degrees
- **(E)** each side 12 cm, each angle 45 degrees
- **(F)** each side 12 cm, each angle 60 degrees

25. An electric iron rated 1.5 kilowatts is used for 40 minutes daily. Over 30 days, how many units of energy does it use, where 1 unit equals 1 kilowatt-hour? [Electric Current & its Effects]

- **(A)** 20 units
- **(B)** 30 units
- **(C)** 60 units
- **(D)** 90 units
- **(E)** 15 units
- **(F)** 22.5 units

26. Paddy (rice) needs water standing on the field, while lentils need well-drained ground. Which soils best suit each crop? [Soil]

- **(A)** sandy-loam for paddy; clayey for lentils
- **(B)** clayey for both crops
- **(C)** sandy for paddy; clayey for lentils
- **(D)** sandy for both crops
- **(E)** clayey for paddy; sandy-loam for lentils
- **(F)** loamy for paddy; clayey for lentils
- **(G)** gravel for paddy; pure sand for lentils

27. The product of a number and 4/7 is exactly 1, so the number is the reciprocal of 4/7. What is the number, and what is three times it? [Fractions & Decimals]

- **(A)** 7/4, and three times it is 21
- **(B)** 7/4, and three times it is 3 1/4
- **(C)** 1, and three times it is 3
- **(D)** 4/7, and three times it is 12/7
- **(E)** 7/4, and three times it is 5 1/4
- **(F)** 4/7, and three times it is 5 1/4
- **(G)** 7/3, and three times it is 7

28. The angles of a triangle are in the ratio 2 : 3 : 5. Find the largest angle, and state whether the triangle is right-angled. [The Triangle & its Properties]

- **(A)** 90 degrees, but it is not right-angled
- **(B)** 100 degrees, and it is right-angled
- **(C)** 90 degrees, and yes it is right-angled
- **(D)** 108 degrees, and it is obtuse
- **(E)** 54 degrees, and it is not right-angled
- **(F)** 90 degrees, and it is obtuse
- **(G)** 72 degrees, and it is right-angled

29. In a circuit symbol the longer line of a cell is the positive terminal. In the wire outside the cell, conventional current flows from ___ to ___ , and adding more cells in series ___ the current. [Electric Current & its Effects]

- **(A)** positive to positive; increases
- **(B)** negative to positive; decreases
- **(C)** positive to negative; does not change
- **(D)** negative to negative; decreases
- **(E)** positive to negative; increases
- **(F)** negative to positive; increases

30. Soil forms when rocks break into tiny pieces over a very long time by the action of sun, water, wind and living things. This breaking is called ___ , and in speed it is mainly a ___ process. [Soil]

- **(A)** leaching; fast
- **(B)** weathering; very fast
- **(C)** erosion; very fast
- **(D)** weathering; very slow
- **(E)** condensation; slow
- **(F)** weathering; instant

31. Petrol costs 102.40 rupees per litre. How much is needed for 2.5 litres, rounded to the nearest rupee? [Fractions & Decimals]

- **(A)** 260 rupees
- **(B)** 256.40 rupees
- **(C)** 205 rupees
- **(D)** 512 rupees
- **(E)** 250 rupees
- **(F)** 256 rupees
- **(G)** 128 rupees

32. An exterior angle of a triangle is (2x + 30) degrees, and its two remote interior angles are x degrees and 70 degrees. Find x and the exterior angle. [The Triangle & its Properties]

- **(A)** x = 50, and the exterior angle = 130 degrees
- **(B)** x = 20, and the exterior angle = 70 degrees
- **(C)** x = 100, and the exterior angle = 230 degrees
- **(D)** x = 40, and the exterior angle = 80 degrees
- **(E)** x = 40, and the exterior angle = 110 degrees
- **(F)** x = 35, and the exterior angle = 100 degrees
- **(G)** x = 40, and the exterior angle = 70 degrees

33. A thin wire heats more than a thick one for the same current. Coil X gives 300 joules per second and coil Y gives 180 joules per second. Running both for 90 seconds, how much MORE heat does X give than Y? [Electric Current & its Effects]

- **(A)** 10800 joules
- **(B)** 43200 joules
- **(C)** 27000 joules
- **(D)** 21600 joules
- **(E)** 16200 joules
- **(F)** 12000 joules

34. A field measures 20 metres by 15 metres. A rain shower adds water at 2 litres for every square metre. How many litres of water fall on the whole field? [Soil]

- **(A)** 1200 litres
- **(B)** 300 litres
- **(C)** 35 litres
- **(D)** 150 litres
- **(E)** 600 litres
- **(F)** 6000 litres

35. Of a sum of money, 1/4 is spent on rent and then 2/5 of the REST is spent on food. What fraction of the whole sum is spent on food? [Fractions & Decimals]

- **(A)** 11/20
- **(B)** 2/3
- **(C)** 2/5
- **(D)** 3/10
- **(E)** 1/10
- **(F)** 1/2
- **(G)** 3/20

36. In a triangle, a straight line from a vertex to the MIDPOINT of the opposite side is called a ___ , while a line from a vertex drawn PERPENDICULAR to the opposite side is called an ___ . [The Triangle & its Properties]

- **(A)** median; bisector
- **(B)** altitude; bisector
- **(C)** altitude; median
- **(D)** median; altitude
- **(E)** bisector; median
- **(F)** median; hypotenuse
- **(G)** side; base

37. A toy needs 3 cells of 1.5 volts. Worn-out cells now give only 1.2 volts each. By how many volts is the toy short of its needed 4.5 volts? [Electric Current & its Effects]

- **(A)** 0.6 volts short
- **(B)** 1.2 volts short
- **(C)** 0 volts short
- **(D)** 1.8 volts short
- **(E)** 0.3 volts short
- **(F)** 0.9 volts short
- **(G)** 1.5 volts short

38. Of 80 millilitres of water poured onto clayey soil, three-fourths stays held in the soil and the rest drains away. How many millilitres drain out? [Soil]

- **(A)** 60 mL
- **(B)** 16 mL
- **(C)** 30 mL
- **(D)** 75 mL
- **(E)** 20 mL
- **(F)** 26.7 mL
- **(G)** 40 mL

39. A wire 8.4 metres long is cut into pieces each 0.6 metre long. How many pieces are there, and is any wire left over? [Fractions & Decimals]

- **(A)** 13 pieces, with 0.6 m left over
- **(B)** 14 pieces, with 0.6 m left over
- **(C)** 12 pieces, with none left over
- **(D)** 7 pieces, with none left over
- **(E)** 14 pieces, with 0.4 m left over
- **(F)** 14 pieces, with none left over

40. A ladder 13 metres long leans against a wall with its foot 5 metres from the base of the wall. How high up the wall does the top of the ladder reach? [The Triangle & its Properties]

- **(A)** 12.5 m
- **(B)** 8 m
- **(C)** 144 m
- **(D)** 18 m
- **(E)** 12 m
- **(F)** 11 m
- **(G)** 9 m

41. A 5-ampere fuse sits in one line. Appliances already drawing 2 amperes and 2.5 amperes are running. What is the largest extra current that can be added before the fuse melts? [Electric Current & its Effects]

- **(A)** 4.5 amperes
- **(B)** 9.5 amperes
- **(C)** 0.5 amperes
- **(D)** 2.5 amperes
- **(E)** 0.25 amperes
- **(F)** 1 ampere
- **(G)** 1.5 amperes

42. Soil R lets 360 mL pass in 30 minutes; soil S lets 200 mL pass in 50 minutes. The ratio of their percolation rates R : S, in simplest form, is ___ . [Soil]

- **(A)** 12 : 4
- **(B)** 9 : 5
- **(C)** 1 : 3
- **(D)** 6 : 1
- **(E)** 3 : 1
- **(F)** 5 : 3
- **(G)** 2 : 1

43. Put these in INCREASING order: 0.5, 0.45, 0.405, 0.054. The correct order is ___ . [Fractions & Decimals]

- **(A)** 0.054, 0.5, 0.45, 0.405
- **(B)** 0.405, 0.45, 0.5, 0.054
- **(C)** 0.054, 0.405, 0.45, 0.5
- **(D)** 0.054, 0.45, 0.405, 0.5
- **(E)** 0.045, 0.054, 0.405, 0.5
- **(F)** 0.5, 0.45, 0.405, 0.054
- **(G)** 0.5, 0.054, 0.45, 0.405

44. Two angles of a triangle are 48 degrees and 67 degrees. Find the third angle and the exterior angle at that third vertex. [The Triangle & its Properties]

- **(A)** third angle 115 degrees; exterior angle 65 degrees
- **(B)** third angle 55 degrees; exterior angle 125 degrees
- **(C)** third angle 65 degrees; exterior angle 65 degrees
- **(D)** third angle 65 degrees; exterior angle 295 degrees
- **(E)** third angle 65 degrees; exterior angle 115 degrees
- **(F)** third angle 65 degrees; exterior angle 130 degrees
- **(G)** third angle 45 degrees; exterior angle 135 degrees

45. Which single change will make an electromagnet stronger WITHOUT changing the battery? [Electric Current & its Effects]

- **(A)** use thinner insulation on the wire
- **(B)** wind more turns of wire on the iron core
- **(C)** use fewer turns of wire
- **(D)** remove the iron core completely
- **(E)** use a wooden core instead of iron
- **(F)** keep the switch closed for longer
- **(G)** make the wire much shorter only

46. Which soil layer is the darkest and richest in humus, and what gives it that dark colour? [Soil]

- **(A)** the topsoil; large sand grains
- **(B)** the subsoil; iron particles
- **(C)** the weathered rock; loose sand
- **(D)** the topsoil; rotted plant and animal matter (humus)
- **(E)** the C-horizon; fresh humus
- **(F)** the bedrock; trapped humus
- **(G)** the bedrock; dissolved minerals

47. Add 3 2/3 and 1 3/4, then subtract 2 5/6 from the sum. What is the result? [Fractions & Decimals]

- **(A)** 3 7/12
- **(B)** 3 1/12
- **(C)** 2 1/12
- **(D)** 2 5/12
- **(E)** 2 7/12
- **(F)** 2 7/6
- **(G)** 1 7/12

48. Which one of these sets of three lengths can actually form a triangle? [The Triangle & its Properties]

- **(A)** 5 cm, 5 cm, 10 cm
- **(B)** 10 cm, 1 cm, 2 cm
- **(C)** 3 cm, 4 cm, 7 cm
- **(D)** 2 cm, 5 cm, 9 cm
- **(E)** 1 cm, 2 cm, 3 cm
- **(F)** 4 cm, 4 cm, 9 cm
- **(G)** 6 cm, 8 cm, 11 cm

49. A geyser of 3 kilowatts runs for 20 minutes each morning. At 5 rupees per unit, what is the weekly (7-day) cost? [Electric Current & its Effects]

- **(A)** 70 rupees
- **(B)** 21 rupees
- **(C)** 35 rupees
- **(D)** 49 rupees
- **(E)** 105 rupees
- **(F)** 7 rupees

50. Water percolates through a soil at 0.5 litre per minute. Express this percolation rate in millilitres per second. [Soil]

- **(A)** 8.3 L/s
- **(B)** 60 mL/s
- **(C)** 500 mL/s
- **(D)** 0.5 mL/s
- **(E)** 0.0083 mL/s
- **(F)** about 8.3 mL/s
- **(G)** 30 mL/s

51. A 0.8-litre bottle is only 35 percent full. How many millilitres of liquid are in it? [Fractions & Decimals]

- **(A)** 560 mL
- **(B)** 320 mL
- **(C)** 2.8 mL
- **(D)** 280 mL
- **(E)** 0.28 mL
- **(F)** 800 mL
- **(G)** 35 mL

52. Each base angle of an isosceles triangle is 53 degrees. Find the vertex angle and name the triangle by its angles. [The Triangle & its Properties]

- **(A)** 74 degrees; an obtuse triangle
- **(B)** 74 degrees; a right triangle
- **(C)** 74 degrees; an acute triangle
- **(D)** 106 degrees; an obtuse triangle
- **(E)** 64 degrees; an acute triangle
- **(F)** 53 degrees; an acute triangle
- **(G)** 127 degrees; an obtuse triangle

53. A compass needle near a wire deflects only when the switch is closed. This shows that ___ , and the deflection reverses when ___ . [Electric Current & its Effects]

- **(A)** the wire is hot; the wire is allowed to cool
- **(B)** the compass is faulty; the compass is shaken
- **(C)** a current produces only heat; the switch is opened
- **(D)** there is static charge; the wire is moved
- **(E)** the cell is weak; the cell is replaced
- **(F)** the wire is magnetic by itself; it is heated
- **(G)** a current produces a magnetic effect; the current direction is reversed

54. Most crops grow best in loamy soil because it ___ , combining the good points of sand and clay. [Soil]

(A) never lets any water drain
(B) cannot hold any air
(C) is made entirely of clay
(D) holds no water at all
(E) holds enough water yet still drains well and is rich in humus
(F) has only the largest particles
(G) is just pure sand

55. A rope is cut in half; one half is cut in half again; then one of those pieces is cut in half once more. The smallest piece is what fraction of the original rope? [Fractions & Decimals]

(A) 1/16
(B) 1/6
(C) 1/3
(D) 1/8
(E) 1/2
(F) 1/4
(G) 3/8

56. A rectangular gate is 8 metres wide and 6 metres tall. A straight brace runs from one corner to the opposite corner. How long is the brace? [The Triangle & its Properties]

(A) 2 m
(B) 10 m
(C) 100 m
(D) 7 m
(E) 14 m
(F) 12 m

57. Five identical bulbs glow in a series circuit. Two of them are unscrewed and removed from their holders. The circuit now ___ because ___ . [Electric Current & its Effects]

(A) stays exactly the same as before
(B) keeps working because the bulbs are in parallel
(C) glows dimmer but still stays on
(D) stops working because the single path is broken
(E) glows brighter because there are fewer bulbs
(F) glows only in the bulbs that remain

58. A nursery needs 0.75 kilogram of soil per pot for 240 pots. Soil is sold in 25-kilogram bags. How many whole bags must be bought? [Soil]

(A) 7.2 bags
(B) 9 bags
(C) 10 bags
(D) 7 bags
(E) 8 bags
(F) 6 bags

59. A car uses 0.08 litre of fuel per kilometre. For a 250-kilometre trip, with fuel at 95.50 rupees per litre, what is the total fuel cost? [Fractions & Decimals]

- **(A)** 764 rupees
- **(B)** 955 rupees
- **(C)** 19100 rupees
- **(D)** 2387.50 rupees
- **(E)** 382 rupees
- **(F)** 191 rupees
- **(G)** 1910 rupees

60. In triangle ABC, angle A is 90 degrees and angle B is 35 degrees. Find angle C, and name the side that lies opposite the right angle. [The Triangle & its Properties]

- **(A)** 55 degrees; side AB is opposite the right angle
- **(B)** 55 degrees; side BC (the hypotenuse) is opposite the right angle
- **(C)** 55 degrees; side AC is opposite the right angle
- **(D)** 65 degrees; side BC is opposite the right angle
- **(E)** 35 degrees; side BC is opposite the right angle
- **(F)** 45 degrees; side BC is opposite the right angle

61. A house used 248.6 units of electricity in March and 191.4 units in April. At 8 rupees per unit, how much LESS was the April bill than the March bill? [Electric Current & its Effects]

- **(A)** 440.00 rupees
- **(B)** 3168.80 rupees
- **(C)** 572.00 rupees
- **(D)** 457.60 rupees
- **(E)** 457.20 rupees
- **(F)** 3520.00 rupees

62. When a dry lump of soil is dropped into water, bubbles rise to the surface. This shows that soil contains ___ , and the bubbling stops when ___ . [Soil]

- **(A)** humus; the lump floats
- **(B)** air; all the trapped air has escaped
- **(C)** salt; the salt has dissolved
- **(D)** nothing; the lump simply sinks
- **(E)** oil; the oil spreads out
- **(F)** water; the soil has dried out
- **(G)** air; the water starts to boil

63. By how much is 7/8 greater than 5/6? Give the answer in lowest terms. [Fractions & Decimals]

- **(A)** 2/14
- **(B)** 1/14
- **(C)** 1/24
- **(D)** 1/2
- **(E)** 2/24
- **(F)** 1/12

64. A triangle has sides 5 cm, 6 cm and 7 cm. A larger triangle of the same shape has its shortest side equal to 15 cm. What is the perimeter of the larger triangle? [The Triangle & its Properties]

- **(A)** 21 cm
- **(B)** 36 cm
- **(C)** 18 cm
- **(D)** 60 cm
- **(E)** 90 cm
- **(F)** 48 cm
- **(G)** 54 cm

65. Two cells are joined so that the positive terminal of one touches the negative of the next; this is a series and their voltages ___ . If instead positive touches positive, the cells ___ . [Electric Current & its Effects]

- **(A)** add up; do nothing at all
- **(B)** halve; double
- **(C)** cancel out; add up
- **(D)** double; double again
- **(E)** add up; add up even more
- **(F)** add up; oppose and partly cancel

66. A pot holds 600 millilitres of water above soil that drains at 8 millilitres per minute. After exactly HALF the water has drained, how many minutes have passed? [Soil]

- **(A)** 4800 minutes
- **(B)** 75 minutes
- **(C)** 37 minutes
- **(D)** 40 minutes
- **(E)** 18.75 minutes
- **(F)** 37.5 minutes
- **(G)** 300 minutes

67. Round 3.14159 to two decimal places, then multiply the rounded value by 100. What do you get? [Fractions & Decimals]

- **(A)** 31.4
- **(B)** 314.16
- **(C)** 3.14
- **(D)** 315
- **(E)** 314
- **(F)** 314.159
- **(G)** 300

68. What is the sum of all three exterior angles of any triangle, taking one exterior angle at each vertex? [The Triangle & its Properties]

- **(A)** it depends on the triangle
- **(B)** 360 degrees
- **(C)** 180 degrees
- **(D)** 270 degrees
- **(E)** 720 degrees
- **(F)** 540 degrees
- **(G)** 90 degrees

69. Bulb P is rated 60 watts and bulb Q is rated 100 watts. Both run for 5 hours. What percent MORE energy does Q use than P? [Electric Current & its Effects]

- **(A)** 20 percent more
- **(B)** about 66.7 percent more
- **(C)** 60 percent more
- **(D)** 40 percent more
- **(E)** 166.7 percent more
- **(F)** 100 percent more

70. Cutting down the forest on a hill speeds up soil loss because ___ , while planting trees and grass ___ the soil. [Soil]

- **(A)** roots push the soil downhill; loosens
- **(B)** wind makes brand-new soil; blows away
- **(C)** the trees eat the soil; create more of
- **(D)** rain adds humus directly; removes
- **(E)** the sun bakes fresh new soil; melts
- **(F)** rain washes away the bare loose topsoil; holds and protects

71. It takes 2/3 of an hour to fill one drum. How many full drums can be filled in 6 hours? [Fractions & Decimals]

- **(A)** 9 drums
- **(B)** 4 drums
- **(C)** 9 1/3 drums
- **(D)** 18 drums
- **(E)** 12 drums
- **(F)** 3 drums
- **(G)** 6 drums

72. A right-angled triangle has legs of 1.5 metres and 2.0 metres. Find the length of the hypotenuse. [The Triangle & its Properties]

- **(A)** 3.5 m
- **(B)** 5 m
- **(C)** 2.25 m
- **(D)** 6.25 m
- **(E)** 1.75 m
- **(F)** 2.0 m
- **(G)** 2.5 m

73. A power board carries devices of 3 amperes, 1.5 amperes and 0.5 ampere. Available fuses are rated 4, 5 and 6 amperes. Which is the smallest fuse that still leaves some room above the total load? [Electric Current & its Effects]

- **(A)** 5.5 amperes
- **(B)** 6 amperes
- **(C)** 4 amperes
- **(D)** 3 amperes
- **(E)** 6.5 amperes
- **(F)** 10 amperes
- **(G)** 5 amperes

74. A loamy soil is 40 percent sand, 40 percent silt, and the rest clay by mass. In a 50-kilogram sample, how many kilograms are clay? [Soil]

(A) 2 kg
(B) 16 kg
(C) 10 kg
(D) 5 kg
(E) 25 kg
(F) 40 kg
(G) 20 kg

75. Three parcels weigh 1.25 kg, 0.8 kg and 2.45 kg. What is their average weight? [Fractions & Decimals]

(A) 1.45 kg
(B) 2.25 kg
(C) 1.5 kg
(D) 1.35 kg
(E) 3 kg
(F) 1.55 kg
(G) 4.5 kg

76. Two angles of a triangle are equal, and the third angle is 30 degrees LESS than each of the equal angles. Find each equal angle. [The Triangle & its Properties]

(A) 50 degrees
(B) 60 degrees
(C) 65 degrees
(D) 70 degrees
(E) 73 degrees
(F) 75 degrees
(G) 80 degrees

77. In an electric bell the hammer strikes the gong again and again because the electromagnet ___ as the circuit is repeatedly made and broken. [Electric Current & its Effects]

(A) stays magnetised forever
(B) switches its magnetic pull on and off
(C) becomes a permanent magnet
(D) gets hotter on every strike
(E) stops the current for good
(F) reverses the battery each time

78. On a hot afternoon, sandy desert soil heats up and cools down much faster than clayey soil mainly because the sandy soil ___ . [Soil]

(A) is always wet underneath
(B) holds a great deal of water
(C) is much darker in colour
(D) holds little water, and water slows down temperature change
(E) contains the most humus
(F) has the smallest particles
(G) reflects no sunlight at all

79. Two-thirds of the students in a class walk to school. Of these walkers, three-fifths are girls. What fraction of ALL the students are girls who walk? [Fractions & Decimals]

- **(A)** 2/3
- **(B)** 2/5
- **(C)** 1/5
- **(D)** 5/8
- **(E)** 6/8
- **(F)** 3/5

80. Sticks of length 5 cm, 12 cm and 13 cm are joined end to end into a triangle. Does it form a right-angled triangle, and why? [The Triangle & its Properties]

- **(A)** yes, because 5 + 12 + 13 = 30
- **(B)** no, the sides are too different in length
- **(C)** yes, because all the lengths are whole numbers
- **(D)** yes, because 5 squared plus 12 squared equals 13 squared
- **(E)** yes, because 5 + 12 is nearly 13
- **(F)** no, it must be an obtuse triangle

81. A family's monthly electricity bill is 1200 rupees at 8 rupees per unit. If the refrigerator alone accounts for one-fourth of the units used, how many units does the fridge use? [Electric Current & its Effects]

- **(A)** 300 units
- **(B)** 37.5 units
- **(C)** 18.75 units
- **(D)** 75 units
- **(E)** 30 units
- **(F)** 150 units
- **(G)** 40 units

82. Water first crosses a sandy layer at 10 mL per minute, then a clayey layer at 2 mL per minute. If 50 millilitres enters, how long does the SLOW clayey layer take to pass that same 50 mL? [Soil]

- **(A)** 5 minutes
- **(B)** 30 minutes
- **(C)** 12.5 minutes
- **(D)** 100 minutes
- **(E)** 10 minutes
- **(F)** 25 minutes

83. You pay with a 500-rupee note for items costing 128.75, 64.50 and 199.99 rupees. How much change should you get back? [Fractions & Decimals]

- **(A)** 96.76 rupees
- **(B)** 106.24 rupees
- **(C)** 393.24 rupees
- **(D)** 116.76 rupees
- **(E)** 106.26 rupees
- **(F)** 106.76 rupees

84. The three angles of a triangle are 80, 60 and 40 degrees. A line is drawn that bisects the 80-degree angle. What are the two new angles created at that vertex? [The Triangle & its Properties]

- **(A)** 40 and 40 degrees
- **(B)** 30 and 50 degrees
- **(C)** 80 and 0 degrees
- **(D)** 45 and 45 degrees
- **(E)** 50 and 30 degrees
- **(F)** 40 and 80 degrees

85. A fuse wire is thin and melts easily. Compared with the thick copper wires of the circuit, the fuse wire should have a ___ melting point so that it ___ first when the current grows too large. [Electric Current & its Effects]

- **(A)** high; cools
- **(B)** high; melts
- **(C)** low; freezes
- **(D)** the same; melts
- **(E)** low; melts
- **(F)** low; cools
- **(G)** very high; melts

86. Humus makes soil fertile because it ___ , and it forms when ___ . [Soil]

- **(A)** releases nutrients as it rots and helps hold water; dead plants and animals decompose
- **(B)** adds extra salt to the soil; the sea dries
- **(C)** is buried plastic waste; factories run
- **(D)** blocks all water from entering; rain falls
- **(E)** kills all the microbes; the sun shines
- **(F)** is a kind of hard rock; lava cools down
- **(G)** is made of pure sand and rock; rocks break

87. Simplify (2/3 + 1/6) divided by (5/6). What is the value? [Fractions & Decimals]

- **(A)** 5/36
- **(B)** 6/5
- **(C)** 5/9
- **(D)** 36/25
- **(E)** 5/6
- **(F)** 1

88. A triangle has sides 4 cm, 9 cm and 7 cm. Which angle is the LARGEST and which is the SMALLEST? [The Triangle & its Properties]

- **(A)** all three angles are equal
- **(B)** largest is opposite the 7 cm side; smallest is opposite the 4 cm side
- **(C)** largest is opposite the 9 cm side; smallest is opposite the 7 cm side
- **(D)** largest is opposite the 9 cm side; smallest is opposite the 4 cm side
- **(E)** largest is opposite the 4 cm side; smallest is opposite the 9 cm side
- **(F)** it cannot be told from the sides

89. Six 1.5-volt cells are placed in series, but two of them are accidentally put in reversed. What is the net voltage across the line? [Electric Current & its Effects]

- **(A)** 1.5 volts
- **(B)** 0 volts
- **(C)** 9 volts
- **(D)** 3 volts
- **(E)** 7.5 volts
- **(F)** 6 volts
- **(G)** 4.5 volts

90. A soil mix has 18 kilograms of sand and 12 kilograms of clay. Written in simplest ratio, and as a fraction of clay in the whole mix, this is ___ . [Soil]

- **(A)** 3 : 2, and clay is 3/5 of the mix
- **(B)** 3 : 2, and clay is 2/3 of the mix
- **(C)** 2 : 3, and clay is 2/5 of the mix
- **(D)** 6 : 4, and clay is 12/18 of the mix
- **(E)** 1 : 1, and clay is half of the mix
- **(F)** 3 : 2, and clay is 2/5 of the mix
- **(G)** 9 : 6, and clay is 1/2 of the mix

91. Write 0.375 as a fraction in lowest terms, and then state its reciprocal. [Fractions & Decimals]

- **(A)** 3/8, and its reciprocal is 8/3
- **(B)** 375/1000, and its reciprocal is 1000/375
- **(C)** 5/8, and its reciprocal is 8/5
- **(D)** 37.5/100, and its reciprocal is 100/37.5
- **(E)** 1/3, and its reciprocal is 3
- **(F)** 3/8, and its reciprocal is 8/5
- **(G)** 3/8, and its reciprocal is 3/8

92. A person walks 8 metres due east, then turns and walks 15 metres due north. How far is she from her starting point in a straight line? [The Triangle & its Properties]

- **(A)** 12 m
- **(B)** 289 m
- **(C)** 17 m
- **(D)** 7 m
- **(E)** 23 m
- **(F)** 11 m

93. An air-conditioner of 1.5 kilowatts runs for 8 hours every night for 30 nights. At 7 rupees per unit, what is the total bill? [Electric Current & its Effects]

- **(A)** 84 rupees
- **(B)** 2520 units
- **(C)** 360 rupees
- **(D)** 5040 rupees
- **(E)** 2520 rupees
- **(F)** 252 rupees

94. A gardener wants water to reach roots at a moderate speed, neither too fast nor too slow. Of soils with percolation rates of 12, 6 and 1 mL per minute, the loamy soil is most likely the one at ___ . [Soil]

- **(A)** more than 12 mL/min
- **(B)** 20 mL/min
- **(C)** both 12 and 1 mL/min
- **(D)** 0 mL/min
- **(E)** 6 mL/min
- **(F)** 12 mL/min
- **(G)** 1 mL/min

95. A printer prints 1.5 pages every second. How many MINUTES will it take to print a 540-page book? [Fractions & Decimals]

- **(A)** 6 seconds
- **(B)** 6 minutes
- **(C)** 360 minutes
- **(D)** 60 minutes
- **(E)** 9 minutes
- **(F)** 3.6 minutes
- **(G)** 12 minutes

96. In a triangle, two of the exterior angles are 130 degrees and 110 degrees. Find all three INTERIOR angles. [The Triangle & its Properties]

- **(A)** 55, 75 and 50 degrees
- **(B)** 40, 60 and 80 degrees
- **(C)** 50, 70 and 120 degrees
- **(D)** 50, 70 and 60 degrees
- **(E)** 60, 60 and 60 degrees
- **(F)** 50, 70 and 100 degrees
- **(G)** 130, 110 and 60 degrees

97. Touching a switch with wet hands is dangerous mainly because ___ , while a fuse protects the wiring by ___ . [Electric Current & its Effects]

- **(A)** water is cold; cooling all the wires
- **(B)** water is magnetic; storing the charge
- **(C)** water lets more current pass through the body; melting and breaking the circuit
- **(D)** water boils instantly; reversing the current
- **(E)** hands turn into magnets; bending the wire
- **(F)** wet hands act as insulators; heating up

98. In a percolation test the water level dropped from the 250 mL mark to the 40 mL mark in 30 minutes. What volume percolated, and what was the rate? [Soil]

- **(A)** 210 mL percolated; 6 mL/min
- **(B)** 170 mL percolated; 5.7 mL/min
- **(C)** 250 mL percolated; 8.3 mL/min
- **(D)** 40 mL percolated; 1.3 mL/min
- **(E)** 290 mL percolated; 9.7 mL/min
- **(F)** 210 mL percolated; 7 mL/min

99. Ravi finished 5/9 of a book while Sara finished 0.6 of the same book. Who read more, and by what fraction of the book? [Fractions & Decimals]

- **(A)** Sara, by 2/9 of the book
- **(B)** Sara, by 1/15 of the book
- **(C)** Ravi, by 1/9 of the book
- **(D)** Sara, by 2/45 of the book
- **(E)** Sara, by 1/45 of the book
- **(F)** they read exactly the same
- **(G)** Ravi, by 2/45 of the book

100. A right-angled triangle has one leg of 20 cm and a hypotenuse of 25 cm. Find the other leg and the perimeter. [The Triangle & its Properties]

- **(A)** other leg 22.5 cm; perimeter 67.5 cm
- **(B)** other leg 15 cm; perimeter 70 cm
- **(C)** other leg 32 cm; perimeter 77 cm
- **(D)** other leg 15 cm; perimeter 45 cm
- **(E)** other leg 15 cm; perimeter 60 cm
- **(F)** other leg 5 cm; perimeter 50 cm
- **(G)** other leg 45 cm; perimeter 90 cm